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Fluid Flow Through Fluidized Beds

Group 4
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 Chemical Engineering Lab I
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Introduction

- The main purpose of this lab is to determine the minimum fluidization velocity needed to create a fluidized bed
- The effect of particle choice and bead height was also studied
 - Different particle diameter
 - Different particle shape
 - Different void fraction
- This lab also served to prove the use of empirical formulas are sufficient for calculations involving fluidized beds

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Theory

- Ergun Equation

$$\frac{\Delta P}{L} = \frac{150 \mu v_{0M}}{\Phi_s^2 d_p^2} \left(\frac{1 - \epsilon_M}{\epsilon_M^3} \right) + \frac{1.75 \rho v_{0M}^2}{\Phi_s d_p} \left(\frac{1 - \epsilon_M}{\epsilon_M^3} \right)$$

Laminar flow term - Neglected if $Re_p \geq 1000$ $Re_p = \frac{\rho v_{0M} d_p}{\mu}$ Turbulent flow term - Neglected if $Re_p < 1$
- Semi-Empirical Equation: $\frac{\Delta P}{L} = g(1 - \epsilon_M)(\rho_p - \rho)$
- Neglect the turbulent term and sub semi-empirical equation into Ergun equation:

$$v_{0M} \approx \frac{g(\rho_p - \rho)}{150\mu} \frac{\epsilon_M^3}{(1 - \epsilon_M)} \Phi_s^2 d_p^2$$

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Apparatus

- Main Equipment
 - Inlet pressure valve
 - Air control regulators
 - Flowmeters
 - Manometer
 - Air distributor
 - Column
 - Disengaging section

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Slide 1

DL1 Good for using the template.
Donglin Li, 9/20/2019

DL13 97/100
Donglin Li, 9/20/2019

Slide 2

DL8 Keep the same font with other titles.
Donglin Li, 9/20/2019

DL9 -1
Donglin Li, 9/20/2019

Slide 3

DL2 Good for presenting the limitations of the equation.
Donglin Li, 9/20/2019

DL3 Should use clearer figures. And this page contains too much information. Could be divided into 2 pages, one for equations, and the other for figures.
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DL4 -1
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Slide 4

DL5 Number the instruments in the figure.
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Material & Supplies

- Materials List
 - Compressed air
 - Silica sand
 - Ion exchange resin beads
 - Vacuums
 - Sieves
 - Graduated cylinder
 - Water

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Procedure

- Prelab Steps
 1. Measure average diameter of sand particles using the sieves provided
 2. Measure the void fraction of both the sand particles and the resin beads using graduated cylinder and water
- Lab Day Steps
 1. Pack 6 in. bed of sand
 2. Incrementally ^{DL12}ase air flow rate and record both the flow rate and pressure drop across the bed until the bed is fluidized
 3. Incrementally decrease air flow rate and record both the flow rate and pressure drop across the bed until the flow is stopped
 4. Repeat 1-3 for a 10 in. packed bed
 5. Repeat 1-4 for ion exchange resin beads

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Experimental Challenges

- Difficulties
 - Precisely controlling ^{DL10} the air velocity from both valves
- Things to watch out for during the experiment
 - Accurately setting the coarse air regulator for total air flow through the bed
 - Correctly packing the bed flat with material
 - Running down the air velocity after achieving minimum fluidization velocity
- Lessons Learned
 - Properly read the entire list procedures prior to starting lab
 - The disengaging section can be removed for easy loading of bed material

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Safety

- High pressure in ^{DL7} valves and flowmeters
- Inhalation of sand dust or resin beads if leaning over column
- Falling off ladder while loading bed with material

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Slide 6

DL12 Good. Could be bigger sizes or divide the text into 2 pages.
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Slide 7

DL10 Too many texts. Could be divided into 2 or more pages.
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Slide 8

DL7 Good for font sizes.
Donglin Li, 9/20/2019